HWA CHONG INSTITUTION
C2 PRELIMINARY EXAMINATION 2009

MATHEMATICS **8863**
Higher 1 **15 September 2009**

3 hours

Additional Materials: Answer Paper
Formulae List (MF15)

INSTRUCTIONS TO CANDIDATES

Write your name and CT group in the spaces provided on the answer paper.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 8 printed pages

[Turn over

Section A : Pure Mathematics [40 marks]

1 Given that $x > 0$, $y = \log_a\left(x^3\right)$ and $z = \log_x a$, show that $yz = 3$.
Hence, find the numerical values of y and z when

$$\log_a(3\log_a x) - \log_a(\log_x a) = \log_a 27.$$ [4]

2 Show that $-x^2 + 2x - 4$ is always negative for all $x \in \mathbb{R}$. [1]

Hence solve the inequality

$$\frac{x^2 - 2x + 3}{-x^2 + 2x - 4} \leq 1.$$ [3]

3 The functions f and g are defined as follows:

$$\mathrm{f} : x \to 1 - \sqrt{(2x - x^2)}, \quad 0 \leq x \leq 2,$$
$$\mathrm{g} : x \to \frac{1}{x}, \quad x \in \mathbb{R}, x \neq 0.$$

(i) Find the largest possible domain in the form of $[0, k]$ such that f^{-1} exists. [1]

(ii) With the new domain, find f^{-1} in a similar form. [4]

(iii) Determine whether gf^{-1} exists, giving a reason for your answer. [2]

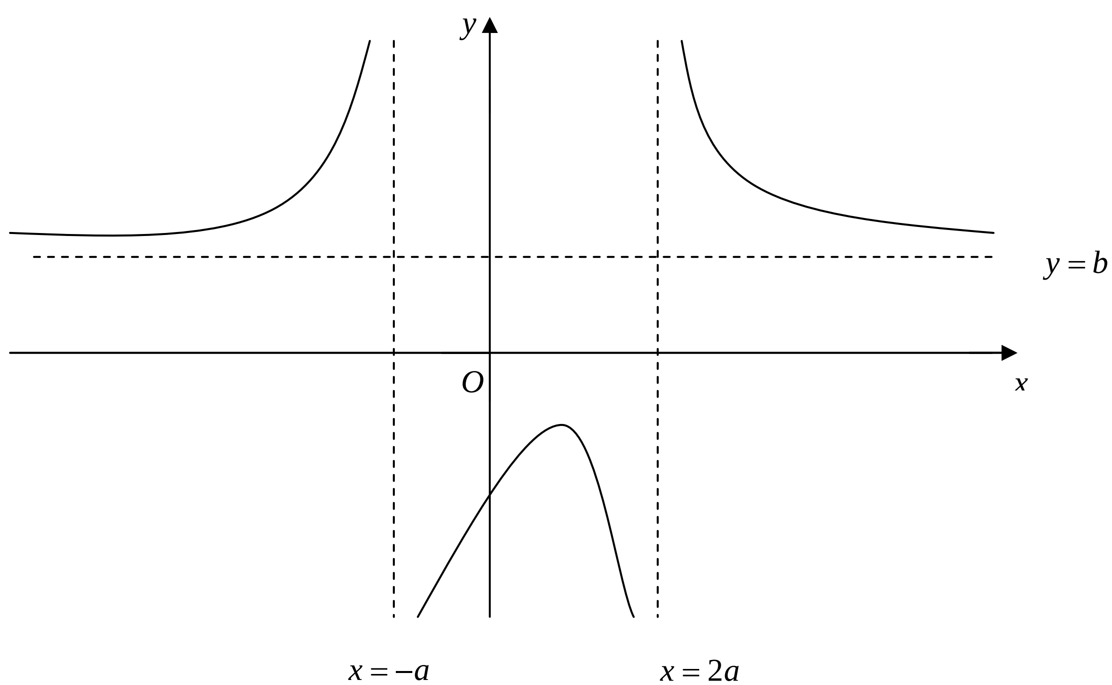
[Turn over

4 (a) Let $f(x)=\dfrac{ax+b}{cx+1}$ and $g(x)=-(x-3)(x+1)^3$, where a, b and c are constants. It is known that $f(0)=g(0)$, $f(3)=g(3)$ and $f(-2)=g(-2)$.

(i) Find the values of a, b and c. [2]

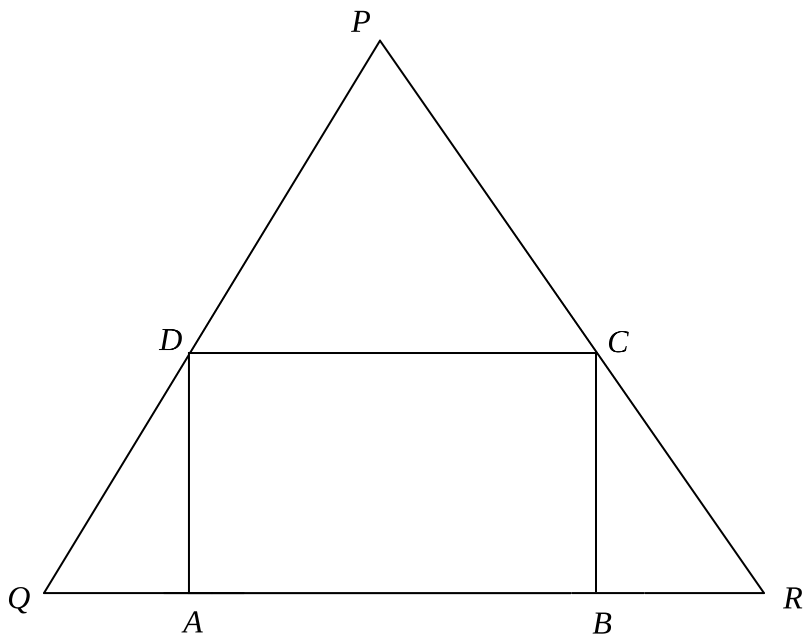
(ii) Sketch the graph of f(x). Stating clearly the equations of asymptotes and the coordinates of any intersections with the axes. [3]

(b) The diagram shows the graph of $y=f(x)$. It has a maximum point $(a,-a)$ and asymptotes $x=-a$, $x=2a$ and $y=b$ where a and b are positive constants. Sketch the graph of $y=f'(x)$, where f' is the gradient function of f, showing the relevant features of the graph. [3]



[Turn over

5 A rectangle $ABCD$ is inscribed in an isosceles triangle PQR as shown in the diagram below.



Given that $PQ = PR = 13\,\text{cm}$, $QR = 10\,\text{cm}$ and $AB = 2x$ cm.

(i) Show that the area, $S\ \text{cm}^2$ of the rectangle is given by $S = ax^2 + bx$ where a and b are constants to be determined. [4]

(ii) Find the value of x for which S is a maximum. [4]

6 Sketch the graph of $y = \ln(x^2 - 3),\ x > \sqrt{3}$ showing the asymptotes and axial intercept(s). [3]

(i) Find the equation of the normal to the curve at the point where $x = 3$, leaving your answer in exact form. [3]

(ii) Find the area of the region bounded by the curve, the x-axis and the normal, giving your answer correct to 4 decimal places. [3]

[Turn over

Section B : Statistics [60 marks]

7 A fitness studio has 1100 members of whom 320 are female. The management wants to obtain the opinion of its members with a view to improving its facilities. A questionnaire is to be sent to the membership. Describe how you could take a systematic random sample of 5% of the membership. State briefly one advantage and disadvantage of this sampling method.

[3]

8 The random variable X has the distribution B(10, 0.15). Find the probability that the mean of a random sample of 50 observations of X is greater than 1.4. [3]

9 In a tea shop, 70% of customers order tea with milk, 20% tea with lemon and 10% tea with neither. Of those taking tea with milk $\frac{3}{5}$ take sugar, of those taking tea with lemon $\frac{1}{4}$ take sugar, and of those taking tea with neither milk nor lemon $\frac{11}{20}$ take sugar. Construct a probability tree showing this information. A customer is chosen at random, find the probability that the customer

(i) takes sugar, [2]

(ii) takes milk or sugar or both, [2]

(iii) takes milk given that the customer takes sugar. [2]

10 Given that A and B are two events such that $P(A \cap B') = \frac{1}{4}$, $P(B \mid A') = \frac{8}{15}$ and $P(B) = \frac{2}{5}$, where A' and B' are complementary events of A and B respectively. Let $P(A) = a$, where $0 < a < 1$.

(i) Express $P(A' \cap B)$ in terms of a, [2]

(ii) Using the fact that $A' \cup B$ is the complementary event of $A \cap B'$, or otherwise, find the value of a, [3]

(iii) Are A and B mutually exclusive? Justify your answer. [2]

11 The length of the metal rods made by a particular manufacturer has mean 15 cm. The setting on the machine is altered and a sample of 50 rods manufactured is found to have the following data:

$$\sum(x-10)=300, \qquad \sum(x-10)^2=5500.$$

(i) Find unbiased estimates of the population mean and variance of the lengths of rods produced by the machine after alternation. [2]

(ii) Test at 3% significance level, whether there is an increase in the mean length of the metal rods produced by this machine? [4]

(iii) State, giving a reason, whether it is necessary for the length of the metal rods to have a normal distribution for the test to be valid. [1]

12 An Internet surfer enters an online competition on the Internet, in which the competitors have to choose the correct answers to a number of questions. There are 4 suggested answers for each question, but the Internet surfer is completely unprepared for the questions and selects an answer at random for each of the questions, so that for each question, the probability of choosing the correct answer is $\frac{1}{4}$. The competition has 50 questions.

(i) Use a suitable approximation to find the probability of the Internet surfer getting more than 17 correct answers. Give your answer to 5 significance figures. [4]

(ii) For a group of 100 Internet surfers, all of whom are unprepared for the competition and select the answer at random for each of the questions, find the probability that at most 10 Internet surfers will each get more than 17 correct answers. [3]

If a competitor is prepared for the above competition, can we still say that the number of correct answers obtained by the competitor follows a binomial distribution? Give a reason to support your answer. [2]

[Turn over

13 Aaron and Heather are about to get married, and decide to buy a car together. Aaron tells Heather that the distance travelled per litre of petrol by the car is dependent on its engine capacity. To prove his point, Aaron shows Heather the following data he collected over the Internet on 9 different types of cars:

	Car 1	Car 2	Car 3	Car 4	Car 5	Car 6	Car 7	Car 8	Car 9
Engine capacity (x cm^3)	2834	989	1358	2415	1474	1991	1608	3786	2195
Distance travelled per litre of petrol (y km/l)	7.2	17.1	15.5	9.2	14.6	11.5	12.6	8.0	10.4

(i) Calculate the product moment correlation between x and y, and explain whether Aaron can claim to Heather that a linear model for his data is appropriate. [2]

On closer inspection on Aaron's data, Heather discovers that one of the data point seems to be incorrect.

(ii) Draw a scatter diagram for the data, and indicate on the diagram the data point that appears to be incorrect by labeling it as P. [2]

Omitting the data point P,

(iii) calculate the equation of the regression line of y on x; [2]

(iv) estimate the value of y when $x = 578$. Comment on the reliability of this estimate; [2]

(v) use a suitable regression line to give an estimate of the engine capacity of a car which can travel 13.2 kilometres per litre of petrol, correct to 4 significance figures. Explain your choice of the regression line. [4]

[Turn over

14 In a supermarket, chicken and ducks are sold by weight. The weight, in kg, of chickens and ducks can be assumed to follow independent normal distributions with means and standard deviations as shown in the table:

	Mean mass	Standard deviation
Chickens	1.8	0.3
Ducks	2.3	0.5

Chickens are sold at \$2.85 per kg and ducks are sold at \$3.20 per kg.

(i) Find the probability of the event that **both** a randomly chosen chicken has a selling price exceeding \$6 and a randomly chosen duck has a selling price exceeding \$8. [3]

(ii) Find the probability that the total selling price of a randomly chosen chicken and a randomly chosen duck is more than \$14. [4]

(iii) Explain why the answer to part (i) is smaller than the answer in part (ii) [1]

The supermarket also sells turkeys by weight, at \$4.50 per kg. It is given that the probability that a randomly chosen turkey has its weight exceeding 11 kg is 0.458, and the probability that a randomly chosen turkey has a selling price less than \$45 is 0.365. If the weight of turkeys can also be assume to follow a normal distribution with mean μ and standard deviation σ, find the values of μ and σ. [5]

The End